

## **Week 5**

# **Understanding under Threat**

Quantum Mechanics and Artificial Intelligence

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# Today's plan

## Part I: Quantum Mechanics

*(lectures 1–2)*

- The framework of QM: states, observables, dynamics
- The measurement problem and the Copenhagen myth
- Interpretations: GRW, Bohm, Everett, ...
- Complementarity: Bohr's response to the crisis
- Bell's theorem and the limits of causal understanding

## Part II: Artificial Intelligence

*(lecture 3)*

- Models and understanding: what philosophy of science says
- The double meaning of “model”
- Sullivan's argument: link uncertainty vs. implementation opacity
- A deeper challenge: is there an inside?

From last week, quantum mechanics is our recurring example of a theory where prediction and understanding seem to come apart.

1. Do you feel that you **understand** quantum mechanics? What would it even mean to understand it?
2. Is “shut up and calculate” a satisfying attitude toward a fundamental theory of nature?

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## Two perspectives today

**First-order** Which interpretation of QM is correct? What is the theory *actually* saying about the world?

**Second-order** There has been no consensus on the foundations of QM for *100 years*. Should we be concerned? What does this tell us about science as a knowledge-producing enterprise?

Both questions bear directly on scientific understanding — which is why De Regt and Dieks chose quantum non-locality as their motivating example.

# The foundational crisis: 1900–1935

- 1900: Planck introduces the quantum of action  $h$
- 1905: Einstein explains the photoelectric effect using photons
- 1913: Bohr's model of the hydrogen atom
- 1925–26: Heisenberg's matrix mechanics; Schrödinger's wave mechanics
- 1927: Born's probability interpretation; Heisenberg's uncertainty principle; Bohr's complementarity
- 1935: The EPR argument — QM must be incomplete

The early pioneers were acutely aware that they did *not* understand what was going on. Bohr thought our ability to understand the physical world was genuinely up for grabs.

## Apparent resolution: the Copenhagen school

- By 1927 a working consensus formed around Bohr's school
- Bohr's key insight: the existence of the **quantum of action** calls for a *revision* of what it means to give an objective, unambiguous description of physical phenomena
  - An unambiguous description must specify the experimental arrangement; the conditions of observation become part of the description itself
  - Classical concepts remain indispensable for describing the apparatus, but cannot all be simultaneously applied to the quantum object
- This leads to **complementarity**: a structural feature of unambiguous description, not a limitation on knowledge
- This is a positive philosophical proposal, not a counsel of despair

# Revival of the crisis

- 1952: Bohm publishes a working hidden-variable interpretation — shattering the apparent consensus
- 1957: Everett proposes the relative-state (many-worlds) interpretation
- 1964: Bell proves his non-locality theorem
- Today: active foundational debate; the cutting edge of physics proceeds without settling the foundations

**The second-order question:** is 100 years without consensus acceptable? Does it reveal something deep about QM, or is it a sociological failure of physics?

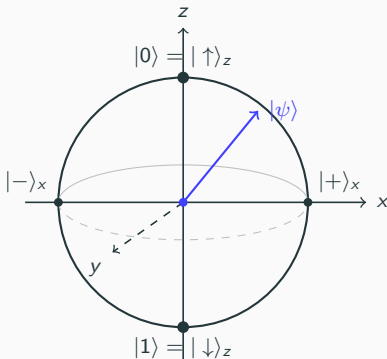


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- The state of a quantum system is a unit vector  $|\psi\rangle$  in a complex Hilbert space  $\mathcal{H}$
- **Superposition principle:** if  $|\psi\rangle$  and  $|\phi\rangle$  are states, so is  $\alpha|\psi\rangle + \beta|\phi\rangle$  (after normalization)
- For a **qubit** (two-level system):  $\mathcal{H} = \mathbb{C}^2$
- Computational basis:  $|0\rangle \equiv |\uparrow\rangle$ ,  $|1\rangle \equiv |\downarrow\rangle$

# The Bloch sphere



Every pure state of a qubit is a point on the sphere. Antipodal points are orthogonal.

# Observables as operators

- Physical quantities: **Hermitian operators**  $A = A^\dagger$
- Possible measurement outcomes: eigenvalues  $\lambda_i$
- Eigenstates  $|a_i\rangle$ : states with definite value  $\lambda_i$

**Spin operators (Pauli matrices  $\times \hbar/2$ ):**

$$S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad S_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

# Superpositions and the puzzle

If the system is in  $|\psi\rangle = \alpha|a_1\rangle + \beta|a_2\rangle$ , observable  $A$  has *no definite value* — yet measurement always yields a definite outcome  $\lambda_1$  or  $\lambda_2$ .

**The puzzle:** what is the system doing between preparation and measurement? Does the question even make sense?

This tension is the engine of every interpretation of QM.

# Uncertainty relations

Non-commuting observables cannot both have definite values:

$$[S_y, S_z] = i\hbar S_x \implies \Delta S_y \cdot \Delta S_z \geq \frac{\hbar}{2} |\langle S_x \rangle|$$

**Spin-z eigenstate decomposed in spin-y basis:**

$$|\uparrow\rangle_z = \frac{1}{\sqrt{2}}|\uparrow\rangle_y + \frac{1}{\sqrt{2}}|\downarrow\rangle_y$$

so  $\Delta S_y = \hbar/2$  is maximal when  $S_z$  is definite.

Conceptually: **Fourier analysis** — narrow in one basis, spread in the conjugate basis.

# Composite systems and entanglement

Composite systems:  $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$ , with  
 $x \otimes (y + z) = (x \otimes y) + (x \otimes z)$ .

Not all states are product states. The **singlet state**

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

cannot be written  $|\psi\rangle_A \otimes |\phi\rangle_B$ : it is **entangled**.

**Discussion (2 min):** what features does a system have if it is entangled with another? What does it mean for its “state”?

# Linear dynamics and the Born rule

Time evolution:  $i\hbar \frac{d}{dt}|\psi\rangle = H|\psi\rangle$ , solution  $|\psi(t)\rangle = U(t)|\psi(0)\rangle$  (**unitary**).

Crucially,  $U$  is **linear**:  $U(\alpha|\psi\rangle + \beta|\phi\rangle) = \alpha U|\psi\rangle + \beta U|\phi\rangle$ . This is exactly what generates the measurement problem.

**Born rule:**

$$\Pr_{\psi}(A = \lambda) = \langle \psi, E^A(\lambda) \psi \rangle$$

For  $|\psi\rangle = \sum_i c_i |a_i\rangle$ :  $\Pr_{\psi}(A = \lambda_i) = |c_i|^2$

**Discussion:** do we *understand* QM if it is just a recipe for computing these probabilities?



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## Definition and derivation

**Ideal measurement of  $A$ :**  $|a_i\rangle \otimes |r\rangle \longrightarrow |a_i\rangle \otimes |y_i\rangle$  (each eigenstate paired with a distinct pointer state — unitary).

Now suppose  $|\psi\rangle = \sum_i c_i |a_i\rangle$ . By **linearity**:

$$\left( \sum_i c_i |a_i\rangle \right) \otimes |r\rangle \longrightarrow \sum_i c_i (|a_i\rangle \otimes |y_i\rangle)$$

The meter ends in an **entangled superposition** — no definite reading. But we *always* observe a definite outcome.

**Contradiction.**

# The collapse postulate

**Standard “solution”:**  $\sum_i c_i |a_i\rangle \longrightarrow |a_k\rangle$  with probability  $|c_k|^2$ .

## Problems:

- Violates unitarity and linearity — the two most fundamental features of the theory
- Creates a two-dynamics theory: Schrödinger evolution *and* collapse
- When does collapse occur? Where is the “Heisenberg cut”?
- The theory gives no definition of “measurement”

# The Copenhagen myth

- Von Neumann (1932) first axiomatized collapse in *Grundlagen der Quantenmechanik*; by the 1950s it was standard textbook orthodoxy
- The label **Copenhagen interpretation** was applied retroactively — Bohr never used it
- In reality there was no single Copenhagen view: Bohr, Heisenberg, Born, Pauli all held different positions
- **HH:** Bohr never claimed the measurement problem is *solved* by collapse — his actual response is philosophically much deeper, and has been systematically misread

# Decoherence: not a solution

**Decoherence:** environmental interaction entangles the system rapidly; the *reduced* state approaches a mixture.

## Why this does not solve the measurement problem:

1. The reduced state is not *uniquely* interpretable as a mixture over definite pointer readings
2. The global state (system + environment) is still a superposition — decoherence is a local approximation, not a collapse

Decoherence explains why interference is hard to observe macroscopically. It does not explain why we observe *one* outcome.

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## Four strategies

1. **Reject unitary dynamics:** add stochastic collapse (*GRW*)
2. **Add hidden variables:** supplement QM with additional ontology (*Bohm*)
3. **Reject uniqueness of outcomes:** all branches are real (*Everett*)
4. **Revise the ontological picture:** resist the demand for a classical reality behind QM (*relational, QBism, pragmatic*)

**GRW** (Ghirardi, Rimini, Weber 1986): add random spontaneous localization events. Free parameters tuned so macroscopic objects localize (no pointer superpositions) while microscopic systems remain unitary. Open: ontology? Relativistic extension?

**Bohm (1952)**: both wavefunction and particle positions are real; the wave guides particles. Particles always have definite positions.

- Guidance equation requires a preferred frame — inconsistent with special relativity
- Violates the spirit of momentum conservation in relativistic interactions



**Everett (1957):** take unitary dynamics completely seriously; no collapse; all branches of the wavefunction are equally real.

We observe one branch because we ourselves are part of the wavefunction — each measurement splits the world.

### Problems:

- *Probability*: the theory is deterministic — where do the Born-rule probabilities come from?
- *Preferred basis*: what selects the “world” branches?
- No-hidden-variables theorems imply the branches cannot themselves be classical

**HH:** these approaches are philosophically sophisticated but may relocate rather than dissolve the problem.

## Less ontological interpretations

**Relational QM** (Rovelli 1996): quantum states are always *relative to an observer* — there is no observer-independent quantum state.

The inspiration is special relativity: velocity has no absolute value, only values relative to a reference frame. Rovelli proposes the same for quantum states. When Alice measures a particle and gets a definite outcome, that outcome is real *relative to Alice*. The particle may still be in superposition relative to Bob, who has not interacted with it. Both descriptions are equally valid. The measurement problem is *dissolved*: collapse is real, but only relative to an observer, never absolute.

**QBism** (Fuchs–Schack): quantum states encode an agent's *beliefs* about future experiences — the wavefunction is epistemic, not ontic.

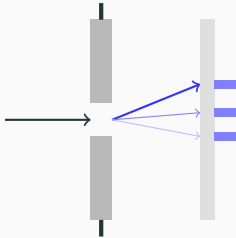
**Pragmatic** (Healey): QM provides a tool for updating credences, not a description of a quantum reality.

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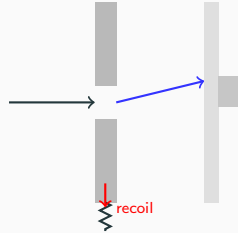
# Bohr's single-slit thought experiment

**A: bolted screen**



position known  $\Rightarrow$  wave picture  $\Rightarrow$   
**interference**

**B: free screen**



momentum known  $\Rightarrow$  particle  
picture  $\Rightarrow$  **no interference**

# Complementarity and visualizability

Two arrangements are **complementary** when they are mutually exclusive and each exhaustive within its domain.

**Spacetime description** position, trajectory, “which path” —  
**visualizable** (anskuelig) picture

**Causal description** momentum, energy, conservation laws —  
dynamical picture

“The viewpoint of complementarity presents itself as a rational generalization of the very ideal of causality.” (Bohr 1948)

This connects directly to De Regt & Dieks: Bohr is *not* saying we cannot understand quantum phenomena. He is proposing a new kind of understanding that does not require a single classical picture.

## Bohr's deeper point

- The quantum of action requires a revision of what counts as an **objective description**:
  - Objectivity does not disappear — but it now requires specifying the experimental conditions as part of any unambiguous statement about a quantum phenomenon
  - We cannot separate “how Nature is” from “the experimental arrangement in which it is observed”
- This is not about abandoning objectivity but about *reconceiving* it for a domain where the interaction between object and instrument cannot be neglected or controlled
- The misreading of Bohr as “shut up and calculate” has distorted the foundations debate for 70 years
- **HH:** Bohr is working in a tradition running from Kant through Høffding: the conditions of unambiguous description are themselves a philosophical achievement

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# EPR, von Neumann, and Kochen–Specker

**EPR (1935):** QM is incomplete — quantum correlations must be explained by predetermined hidden variables  $\lambda$ .

**Von Neumann (1932):** no hidden-variable theory can reproduce QM given additivity  $\mathbb{E}_\lambda(A + B) = \mathbb{E}_\lambda(A) + \mathbb{E}_\lambda(B)$  for non-commuting  $A, B$ . Bell (1966): this assumption is physically unmotivated.

**Kochen–Specker (1967):** no assignment  $v(A) \in \text{spec}(A)$  to all observables can satisfy functional consistency  $v(f(A)) = f(v(A))$  and the sum rule for commuting pairs. **Value definiteness** is impossible.

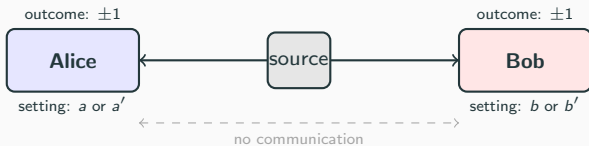
Kochen–Specker is more compelling than von Neumann: the assumptions are operationally motivated.



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# The CHSH scenario



The entangled pair is measured far apart. Each party independently chooses a setting and records a  $\pm 1$  outcome.

## Local hidden variables and the CHSH inequality

Suppose outcomes are determined by  $\lambda \sim \rho(\lambda)$ , with **reality** ( $A, B$  definite) and **locality** (Alice independent of Bob's setting). Define the **correlation** between outcomes:

$$E(a, b) = \int A(a, \lambda) B(b, \lambda) \rho(\lambda) d\lambda$$

i.e. the expectation value of the product of Alice's and Bob's outcomes ( $\pm 1$  each) for settings  $a$  and  $b$ . Perfect correlation:  $E = +1$ ; perfect anti-correlation:  $E = -1$ ; no correlation:  $E = 0$ .

Under local hidden variables:

$$|E(a, b) - E(a, b') + E(a', b) + E(a', b')| \leq 2$$

*Proof sketch:* for each  $\lambda$ , the combination

$A(a)[B(b) - B(b')] + A(a')[B(b) + B(b')]$  takes values in  $\{-2, 0, +2\}$ .

# Quantum violation

For  $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$ :  $E(a, b) = -\cos(\theta_a - \theta_b)$

Choose  $a = 0^\circ$ ,  $a' = 90^\circ$ ,  $b = 45^\circ$ ,  $b' = 135^\circ$ :

$$E(a, b) = -\cos(45^\circ) = -\frac{1}{\sqrt{2}}$$

$$E(a, b') = -\cos(135^\circ) = +\frac{1}{\sqrt{2}}$$

$$E(a', b) = -\cos(45^\circ) = -\frac{1}{\sqrt{2}}$$

$$E(a', b') = -\cos(45^\circ) = -\frac{1}{\sqrt{2}}$$

$$S = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -\frac{4}{\sqrt{2}} = -2\sqrt{2}$$

$$\Rightarrow \boxed{|S| = 2\sqrt{2} \approx 2.83 > 2}$$

# The Jarrett analysis and what Bell means

Jarrett (1984): local hidden variables = two conditions: **PI (parameter independence)** Alice's outcome does not depend on Bob's *setting*

**OI (outcome independence)** Alice's outcome does not depend on Bob's *outcome*

Bell violation  $\Leftrightarrow$  at least one of PI or OI fails.

- **Orthodox (collapse):** OI fails — collapse makes Alice's probability depend on Bob's *outcome*
- **Bohm:** PI fails — the quantum potential makes Alice's probability depend on Bob's *setting*
- **Everett:** unclear — no unique outcomes, so the framework is hard to apply. (Everettians typically *claim* the theory is local — but making this precise is notoriously difficult.)
- **Retrocausal models:** PI fails — future settings influence

## Taking stock — and a question

- QM achieves extraordinary predictive precision: the anomalous magnetic moment of the electron agrees with experiment to 12 significant figures
- Yet the theory is arguably *inconsistent as stated* (the measurement problem), and after 100 years there is still no consensus on how to resolve it
- Each interpretation pays a serious price: non-locality, many worlds, anti-realism, new fundamental dynamics

Some physicists say: “QM works, so who cares if we don’t understand it?” If we accept this, we have weakened our ground against:

*“AlphaFold predicts protein structures better than any human theory — so who cares if we don’t understand it?”*

**HH:** The goal of science is not mere prediction but understanding<sup>38</sup> / 61

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# Graduates Boo Commencement Speech About A.I.

Humanities students made their displeasure known at the University of Central Florida.

Listen · 5:54 min

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Students at the University of Central Florida booed a commencement speaker after she said that “artificial intelligence is the next industrial revolution.” University of Central Florida



AlphaFold (2022) predicted the 3D structure of virtually every known protein — a problem biologists had worked on for 50 years. AlphaFold is a deep neural network trained on known protein structures.

1. Did AlphaFold help us **understand** protein folding? Or did it only help us **predict** protein structures?
2. Is there a meaningful difference between a neural network that predicts and a model that explains?

## Two perspectives on AI in science

**Theoretical** Science has always used tools to extend understanding. But AI tools seem borderline *intelligent* themselves. Does that change what the task of science *is*?

**Practical** Science produces two kinds of good: **knowledge** about the world, and **power** over the world. AI threatens to tip the balance away from knowledge.

What kind of science do we want to do?

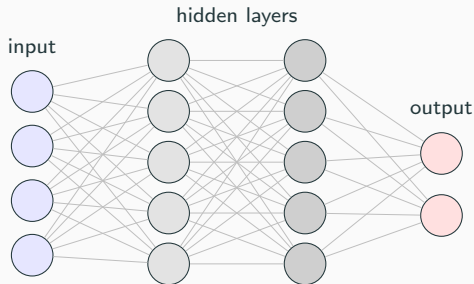
## A fact about AI: opacity

Deep Neural Networks (DNNs) — including the transformer models behind AlphaFold and large language models — share one crucial feature:

### Even the experts who build them do not know how they work

- There is a fundamental **mismatch** between human reasoning and the path a DNN takes from inputs to outputs
- A DNN learns by adjusting billions of numerical weights through an optimization process; the resulting weights are not human-readable
- Joel Dudley (co-creator of the Deep Patient model): “We can build these models, but we don’t know how they work”

# What a DNN looks like inside



each edge carries a weight  $w_{ij} \in \mathbb{R}$ , learned by gradient descent — not designed to represent any physical quantity

The “knowledge” lives in the weights. These are the output of gradient descent over training data — not designed to represent any physical mechanism or causal structure.

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## Models and understanding

Last week we discussed DN, causal-nomological, and unificationist accounts of explanation. Despite their differences, they share one assumption:

*A genuine explanation says something about what is going on in the world — it does not merely predict.*

Contemporary philosophy of science stresses that explanations are typically underwritten by **models**: representations with interpretable content.

Such a model is not a black box for computing predictions. It has an *inside* that scientists can inhabit, reason about, and transfer to new situations.

## Two examples

**Kinetic theory** gas = molecules in random motion; explains temperature, pressure, and effusion rates by appealing to real physical processes.

**Schwarzschild metric** explains why light bends *twice* the Newtonian prediction, because spatial curvature is a genuine feature of the geometry near a mass.

**Pointer to the literature:** van Fraassen, *Scientific Representation: Paradoxes of Perspective* (Oxford, 2008) — the central philosophical study of how models represent their target systems.

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*“Are scientists trading understanding for some other epistemic or pragmatic good when they choose an opaque and complex machine learning model?”*

(Sullivan 2022, p. 110)

**Discuss:** do you think scientists are making such a trade? Is that acceptable?

# The double meaning of “model”

One terminological trap: the word **model** does double duty.

## Sense 1 — scientific model

Interpretable content: states a mechanism; has an inside that scientists can inhabit; satisfies CIT.

*Examples: kinetic theory, Schwarzschild, BCS*

## Sense 2 — ML model

A parameterized function *fitted to data by optimization*: weights set by gradient descent, not designed to represent anything.

*Examples: AlphaFold, GPT, Deep Patient*

## Why the ambiguity matters

The word “model” in ML descends from statistics, where it *did* once have Sense 1 content (a Gaussian model, a linear model makes readable claims about a data-generating process). Neural networks inherited the term while quietly abandoning the interpretability.

Sullivan’s article uses “model” in both senses without clearly distinguishing them. She asks whether a **Sense 2** model can provide the understanding we normally expect from a **Sense 1** model.

This is the right question — but answering it requires acknowledging that there is a genuine gap to be bridged, not just a terminological overlap.

## Sullivan's main claim

Sullivan's answer to her own question: it is *not* the complexity or opacity of a DNN that limits the understanding it can provide.

The real culprit is **link uncertainty**:

*"It is not implementation black-boxing that gets in the way of understanding, it is link uncertainty."*

(Sullivan 2022, p. 116)

- Black-boxing implementation details does *not* undermine understanding in general — this is commonplace in science
- What blocks understanding is the absence of scientific evidence linking the model's outputs to real causal structure in the world

**Discuss:** do you think Sullivan is right about this? Can you give examples of acceptable and unacceptable black-boxing from physics?

# Black boxes

A **black box**: a system whose inputs and outputs are known but whose internal workings are not accessible.

**Implementation black-boxing** is ubiquitous in science — and usually *unproblematic*:

- A climate model's factorial subroutine may be iterative or recursive — it makes no difference to the climatology
- Schelling's model gives the same result on a computer, a real board, or a Go board

**A familiar case**: many physicists treat QM itself as a **quantum statistical algorithm** — preparations go in, measurement statistics come out, do not ask what is happening inside. Part I has been arguing this attitude is unsatisfactory.

# Black boxes in machine learning

DNNs are black boxes at a *deeper* level: even the designers cannot identify which features the model has latched onto.

The opacity goes beyond implementation details:

- Weights are set by optimization, not by design
- The modeller cannot predict which data features will be most salient
- Indirect probing (e.g. saliency maps) gives only approximate, high-level information

**Sullivan's claim:** even this deeper opacity does not block understanding. The real obstacle is something else.

# The Schelling analogy

Sullivan's central example: Schelling's **checkerboard model** of residential segregation.

- Simple simulation: two types of agent, each moves if  $> 70\%$  of neighbours differ; result: a segregated board
- Provides *how-possibly* understanding when the semantic mapping (coin = agent, move = choice) is read off
- Becomes *how-actually* understanding only when empirical evidence connects the model to real populations

Sullivan's claim: the same structure applies to DNNs.

Implementation opacity is not the obstacle — link uncertainty is.

**HH:** But notice the key difference. With Schelling, the semantic mapping was *built in by the modeller from the start*. With Deep Patient, there is no such mapping. The very question of what the nodes and weights *refer to* is unanswered — and it is not obvious it is even askable.<sup>55</sup> / 61

## Sullivan's optimism and a challenge

*"Once the link uncertainty is resolved, the deep patient model is able to explain and enable understanding of disease development."*

(Sullivan 2022, p. 125)

**HH:** How would the link uncertainty ever be resolved in this case?

- With Schelling: we look for evidence that individual preferences actually drive segregation — this is a clearly statable empirical question
- With Deep Patient: what physical objects or processes would the learned weights be *linked to*?
- The weights emerged from optimization over medical records. There is no design that maps them onto causal factors in disease. The link question cannot be stated in the same form.



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## The internal/external distinction

When a physicist derives a result, her brain is — at the physical level — just as opaque as a DNN.

If you traced the neural mechanisms, you would not see the reasoning. To understand why she gets the right answers, you have to get inside the **structure of the reasoning itself**.

**Leibniz's mill** (*Monadologie*, §17): even if the brain were enlarged to the size of a mill, and you could walk around inspecting all its machinery, you would never encounter *thought* anywhere in the mechanism.

Sullivan asks: how opaque is the implementation?

But there is a prior question:

**Is there an internal perspective to be had at all?**

- With the physicist: yes — her reasoning has a logical structure you can inhabit
- With a DNN: *open question*

## Van Fraassen: a model must be a “way the world could be”

Van Fraassen is the arch empiricist: he does not ask that our theories be *true*, only *empirically adequate*.

Yet even van Fraassen would not be satisfied with Deep Patient as a final scientific result.

His **constructive empiricism** is a *semantic* empiricism, not an eliminativist one. A scientific theory presents models — structures interpretable as *ways the world might be*. The scientist may be agnostic about truth, but she must be able to say: this is a *possible description of reality*.

A DNN does not present a way the world could be. It presents a function from inputs to outputs. The question of truth or falsity about disease mechanisms does not even arise.

## For reflection

1. Should scientists rest content with having something as predictive as Deep Patient? Or does science have a further goal that remains unfulfilled?
2. Does it matter whether we are talking about *applied* science (predicting disease, protein structure) vs. *fundamental* science (understanding the laws of nature)?
3. We saw in Part I that QM proceeds without consensus on what it means. Does this weaken or strengthen the case for demanding understanding from AI?
4. Intimation: the problem of understanding another human being — and why that seems different from understanding a DNN — points toward questions that philosophy has been wrestling with for centuries.

